

Receive and Check In a Carton SOP

One episode receives one carton. The tape strip is cut along the seam, all four flaps are folded down, the packing slip is laid face up on the desk, and the three items come out of the carton into one row on the desk to the left of the carton, left to right in slip line order. The carton is empty, all four flaps are still down, and the cutter is back on its rest when the episode ends.

The order never changes: cut, flaps, slip, then the three items in slip line order 1 to 3. Nothing is lifted out of the carton until the slip is face up on the desk, and the line is read before each item is lifted. Every read is made by the **right gripper**, held over the line on the slip. The reads are the count, so the slip is not read again at the end.

The carton sits on the bare desk with nothing holding it, so the gripper that is not working anchors it. The anchor goes either against the outside of the carton wall on its own side or down on a flap that is still closed on the top, whichever holds better. The carton is never lifted, slid, tipped, or turned, and an anchor on the top stays clear of the seam and of the blade's path while the other gripper is cutting.

Every item is set down in the row by the **left gripper**. Lines 1 and 2 are packed in the carton's left half, and the left gripper lifts them straight out and stages them itself. Line 3 is packed in the carton's right half, too far for the left arm to reach cleanly, so the **right gripper** lifts that one and passes it to the left gripper in the air, and the left gripper stages it. The right gripper never sets an item down on the desk and never enters the row.

The right gripper works the safety cutter, the right side flap, the near end flap, the slip, every read, and the lift of line 3 up to the handoff. The left gripper works the left side flap, the far end flap, the carton's left half, and the whole row, and it anchors the carton for the cut. Only the right gripper ever touches the cutter or the slip.

The blade only ever travels along the tape strip, no flap is torn open against tape that is still joined, nothing is turned in the air, no item is pulled out from between its neighbors, and no gripper opens anywhere but over the spot the item is going into. The pass is the one exception: the right gripper opens on the line 3 item in the air, and only once the left gripper is already closed on it.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole desk: the carton at back center, the clear band of desk to the left of the carton where the row is built, the handoff point in front of the carton, the slip spot on the right side of the desk, and the cutter rest at the front right. The slip lines are readable in frame, so a gripper held over a line can be seen on the line it is reading.
3. The whole tape seam is in frame end to end, and the carton interior is visible from above so every lift can be seen leaving it.
4. Both arms are at home with grippers open.
5. The desk is bare apart from the carton and the cutter. There are no marks, trays, bins, or fixtures on it.
6. Nothing obstructs the paths between the carton, the handoff point, and the band of desk to the left of the carton. The left arm reaches the carton's left wall, the carton's left half, the handoff point, and every spot in the row. The right arm reaches the carton's right wall, the whole tape seam end to end including its left end, the carton's right half, the handoff point, the slip spot, and the cutter rest. Both grippers reach the handoff point holding an item level, without either arm leaning out.

Materials checklist

1. One carton sitting square on the desk at back center, base flat on the desk, its four top flaps closed and its two end flaps meeting along the seam, with clear desk on all four sides of it.
2. The carton has **four top flaps**: two **end flaps** hinged along the front and back top edges, lying on top and meeting edge to edge along a **seam** that runs left to right across the top, and two **side flaps** hinged along the left and right top edges, folded in flat underneath them. The end flap at the front is the **near end flap** and the one at the back is the **far end flap**.
3. One **tape strip** lies flat along the seam, covering it end to end, with no tape carried down the front, back, or side walls.
4. One **safety cutter** lies flat on the desk at the front right, head pointing away from the front edge, with a recessed hook blade seated and its depth set so a flat draw parts tape only.
5. One packing slip, printed, stiff enough to keep its shape when pinched by an edge, listing three numbered lines with one item per line, printed large enough to read in frame. It lies face up on top of the contents in the carton's **right half**, where the right gripper takes it.
6. **Three items** lie in the carton in one layer, in a single row running left to right in slip line order, with a clear gap between them so a gripper can close on one without touching its neighbors. Lines 1 and 2 sit in the carton's left half and line 3 sits in the right half. Each item is small enough for one gripper to close on it at its middle, heavy enough to stay put once it is set down, and stable on the bare desk in the orientation it is packed in. The **line 3** item is also square enough at its middle to be held by one gripper across one pair of opposite faces

while the other gripper closes on the other pair at the same time, so it can be passed in the air without either gripper letting go. Nothing rests on top of anything else.

7. The desk to the left of the carton is clear, with room for three items in a row between the carton's left wall and the left edge of the desk. The desk in front of the carton is clear under the handoff point.

Workspace layout

Every zone is a position on the bare desk. Nothing is marked and nothing holds anything, so each position is held by where the last thing was put.

- **Carton:** at back center, centered between the two arms, close enough that both arms reach its top, including the near and far top edges, without leaning out. It sits on the desk for the whole episode.
- **Row band:** the clear band of desk to the left of the carton, between the carton's left wall and the left edge of the desk, clear of the left side flap once it is folded down. The three items are built into one row here, left to right in slip line order: **line 1 lands at the outer left end** and the row grows back toward the carton, so line 3 finishes nearest the carton's left wall with a clear gap off it. The whole band is in the left arm's reach.
- **Handoff point:** over the clear desk in front of the carton, centered between the two arms, at a height where both grippers hold the item level without either arm leaning out. The line 3 item is passed from the right gripper to the left gripper here, and nothing else ever happens here. It is clear of the carton, the row band, the slip spot, and the cutter rest, so nothing is ever above any of them during a pass.
- **Slip spot:** on the right side of the desk, between the carton and the cutter rest, a clear gap off the carton's right wall and clear of the row band, the handoff point, and the cutter. The slip lies face up here, square to the front edge, in the right arm's reach so the right gripper can read it without leaning across the carton or the row.
- **Cutter rest:** the front right corner of the desk, on the far side of the carton from the row band and clear of the slip spot. The cutter lies flat here, head away from the front edge, before and after the cut.

Lines 1 and 2 are packed in the carton's left half, and the left gripper lifts them straight out and stages them. Line 3 and the slip are packed in the right half. The slip is the right gripper's for the whole episode, and line 3 is lifted by the right gripper, passed to the left gripper at the handoff point, and staged by the left gripper like the other two. No carried item passes over the carton, over the slip, or over an item already in the row, and an item crosses the midline only in the pass at the handoff point. The right gripper travels to the slip spot empty and closed for every read, and it stays on its own side of the desk doing it.

Vocabulary

- **Arm assignment:** the right arm works the cutter, the tape seam, the right side flap, the near end flap, the carton's right half, the slip, every read, and the lift of the line 3 item up to the handoff point. The left arm works the left side flap, the far end flap, the carton's left half, the handoff point, and the whole row: every item that ends up on the desk is put there by the left gripper. Either arm anchors on the wall on its own side. Neither arm takes the other's zones. The left gripper never touches the slip or the cutter, and the right gripper never reaches into the carton's left half, never enters the row band, and never sets an item down on the desk.
- **Seam:** the line where the two closed end flaps meet, running left to right across the top of the carton. It has a **right end** and a **left end**. Only the end flaps meet at the seam; the two side flaps lie underneath them and are not taped.
- **End flaps:** the two top flaps hinged on the front and back walls. They are on top, they carry the seam and the tape, and they are the first two opened. The **near end flap** is at the front, the **far end flap** at the back.
- **Side flaps:** the two top flaps hinged on the left and right walls, lying flat under the end flaps with their free edges toward the middle of the carton. They are opened after both end flaps are down.
- **Tape strip:** the single strip covering the seam. It is the only thing the blade ever touches.
- **Safety cutter:** the tool on the cutter rest. Its **handle** has a flat grip section and its **head** carries the recessed blade. A gripper closes on the handle and never on the head.
- **Anchor:** the closed gripper holding the carton still on the bare desk while the other gripper works. It takes either of two forms, and the operator uses whichever holds better. A **wall anchor** is pressed onto the outside of the carton wall on its own side, level with the middle of the wall. A **top anchor** is pressed down onto a flap that is still closed on the top of the carton, on its own side of the midline, clear of the seam and of the flap the other gripper is working. If no closed flap is left free on its own side, the anchoring gripper uses the wall. An anchor never travels, never presses on the contents, never reaches across the open top, and lifts clear only after the working gripper is off the carton. Nothing else holds the carton, so every cut and every flap fold is anchored.
- **Draw:** the right gripper carrying the cutter along the seam, right end to left end, in one pass with the sole of the head flat on the carton top. The pass does not have to be unbroken: if the blade catches or drags, the draw slows or stops with the head still down on the strip and then carries on from where it stopped. What it never does is come off the strip and go back on. The cutter is the only thing that crosses the midline, and only above the closed carton top during the draw.
- **Flap down:** a flap folded out over its hinge, creased, and left lying flat against the outside of its own wall, leaving the top opening clear. All four flaps go down, each against its own wall, so the opening is clear front to back and side to side.

- **Slip line:** a numbered line on the packing slip. The item on line N is the Nth item placed in the row, and the line is read before the item is lifted.
- **Read a line:** the **right gripper**, closed and empty, travels to the slip spot, stops with its tip just above the line being read and clear of the slip face, and holds still there for 2 seconds so the line it is on is visible in frame. It then lifts clear. A read never touches, presses, slides, or covers the slip, and no item is ever in the gripper during a read.
- **Count check:** the three reads themselves. Line N is read, the item on it goes to position N in the row, and the row that results is the count. There is no second pass over the slip at the end.
- **Item:** one of the three things packed in the carton. Items are not named in this SOP and are not sorted, opened, read, or inspected. Each one is taken at its grasp, moved once, and set down in the row.
- **Item grasp:** the gripper comes down on the item at its middle and closes across the two opposite faces that let it be lifted level, clear of any lid, cap, corner, loose edge, or moving part. Each item has one grasp and it is taken the same way every episode.
- **Cross grasp:** the second hold on the **line 3** item only, at the same middle but across the other pair of opposite faces, at right angles to the grasp the right gripper is already holding. It is the left gripper's hold in the pass and it is used nowhere else.
- **Pass:** the line 3 item going from the right gripper to the left gripper at the **handoff point**, held level throughout and never let go by both grippers at once. The right gripper carries the item to the handoff point and holds still. The left gripper comes in open and empty and closes on the **cross grasp**. Only once the left gripper is closed does the right gripper open and lift clear. The item is not turned, tilted, shifted, or regripped during the pass, and it never touches the desk between the carton and the row.
- **Row:** the three items standing on the bare desk in one line to the left of the carton, built left to right in slip line order from the outer left end back toward the carton, roughly in line with each other, with a clear gap between neighbors and nothing touching. Every item in it is set down by the left gripper. The row order is the count, so an item out of order is a miscount.
- **Next spot:** the clear desk just to the right of the last item placed, that is, one step back toward the carton, far enough over that the two do not touch. The first item lands at the outer left end of the row band and sets where the row starts, and the last one finishes with a clear gap off the carton's left wall.
- **Lift:** taking an item straight down onto its grasp, closing, and raising it straight up until there is daylight under it, before anything moves sideways.
- **Stage:** carrying an item level to its next spot, in the orientation it was packed in, lowering it straight down until it is resting flat on the desk, and releasing once it is resting.
- **Settled:** the item stays put for 2 seconds after the gripper lifts clear, with nothing rocking, sliding, or falling over.

- **Release point:** over the next spot, the slip spot, or the cutter rest the thing in the gripper is going to, once it is already resting. A gripper opens nowhere else, with one exception: the right gripper opens on the line 3 item at the handoff point, and only after the left gripper has closed on it.

Steps

Steps 1 to 4 are one carton. Every item is lifted straight up out of the carton and carried in the orientation it will be set down in.

Step 1: Cut the tape along the seam

Goal: the tape strip parted along the whole seam, the carton still square where it started, and the cutter back on its rest.

1.1 Anchor the carton and take the cutter

- The **left gripper**, closed, anchors the carton and holds it there for the whole step: either against the **left wall**, or pressed down on the carton top on the near or far flap, a clear margin back from the **seam** so the whole draw passes beside it.
- The **right gripper** comes down onto the flat of the **handle** on the cutter rest, closes across it, and lifts the cutter straight up.

1.2 Draw the cutter along the seam

- Bring the head down onto the **right end** of the tape strip until the sole of the head sits flat on the carton top and the blade is through the tape.
- Draw the cutter along the seam, right end to left end in one pass, keeping the head flat and on the strip, until the blade passes the left end of the tape.
- If the blade catches or the cutter stops moving, hold the head where it is, flat on the strip, and carry on along the seam from that point. A stall is not a restart, so the head does not come off the strip and there is no second entry into the tape.

1.3 Return the cutter to its rest

- Lift the cutter straight up clear of the carton, carry it back to the **cutter rest**, lower it flat onto the desk head away from the front edge, and release.
- The **left gripper** lifts clear once the cutter is on its rest.

Nothing but the anchor holds the carton, so the draw runs toward the anchor on the left and pushes the carton into the gripper holding it rather than away from it. A top anchor holds the carton down as well as still, but it has to sit far enough off the seam that the head and the blade run past it with room to spare. The right arm passes over the midline to finish the draw, but only above the closed carton top, and never over the row or the slip. The blade is set to tape depth and the head stays flat, so the cut parts the tape without reaching the contents. Pausing

partway is fine and a slow stretch where the blade drags is fine, but never lift the head off the strip and set it down again partway along the seam.

Check: the tape is parted along the whole seam, the two end flaps are free of each other, the carton sits square where it started and has not slid or turned, and the cutter lies on its rest. If the tape is still joined anywhere, anchor again and draw the joined part with the same flat motion. Never pull the end flaps apart against tape that is still joined.

Step 2: Fold all four flaps down

Goal: all four flaps folded down against their own outside walls with the top opening clear.

The two end flaps come off first because they carry the tape and lie on top of the side flaps. Each flap is opened by one gripper while the other anchors the carton on its own side, on the wall or on a flap that is still closed there.

2.1 Fold the far end flap down

- The **right gripper** anchors the carton on its **right side**, wall or closed flap.
- The **left gripper** pinches the free edge of the **far end flap** at its middle, lifts it up and out over its hinge in one continuous motion, and lays it flat against the carton's back outside wall.
- Press it there for 2 seconds so the crease sets, release, and lift clear.

2.2 Fold the near end flap down

- Swap the roles: the **left gripper** anchors the carton on its **left side**, wall or closed flap.
- The **right gripper** takes the free edge of the **near end flap** the same way and lays it flat against the carton's front outside wall, presses for 2 seconds, releases, and lifts clear.

2.3 Fold the right side flap down

- The **left gripper** anchors the carton on its **left side**, wall or closed flap.
- The **right gripper** pinches the free edge of the **right side flap** at its middle, lifts it up and out over its hinge, and lays it flat against the carton's right outside wall.
- Press it there for 2 seconds, release, and lift clear.

2.4 Fold the left side flap down

- The **right gripper** anchors the carton on its **right side**, wall or closed flap.
- The **left gripper** takes the free edge of the **left side flap** the same way and lays it flat against the carton's left outside wall, presses for 2 seconds, releases, and lifts clear.
- Both grippers lift clear of the carton.

The end flaps sit on the midline, so either gripper can take one: the left gripper takes the far flap and the right gripper takes the near flap, and the gripper that is not folding always anchors on its own side, on the wall or on a flap that is still closed there. Each side flap is then opened by

the gripper on its own side and anchored by the other, so the fold pulls against a held carton instead of dragging it across the desk. By the last fold there is no closed flap left on the right, so that anchor goes on the wall. Never open a side flap while an end flap is still lying over it.

Check: all four flaps lie down against their own outside walls, none is standing up, leaning back over the opening, or still tucked in flat over the contents, the top opening is clear front to back and side to side, the packing slip is in view, and the carton has not moved. If a flap springs back over the opening, anchor again, take its free edge, and crease it down again.

Step 3: Lay the packing slip on the desk

Goal: the slip lying face up at the slip spot and readable, with all three items uncovered.

- The **right gripper** pinches the near edge of the slip at its middle and lifts it straight up out of the carton's right half, level and face up.
- Carry it level to the **slip spot** on the right side of the desk, lower it flat onto the desk square to the front edge, and release once it is resting.
- The **left gripper** stays clear of the slip and of the right side of the desk for the whole step.
- The slip stays where it lands for the rest of the episode. It is read from there and never picked up, moved, or touched again.

Check: the slip lies flat and face up at the slip spot, its three lines readable in frame, clear of the carton, the handoff point, and the cutter rest, and in the right arm's reach. All three items in the carton are uncovered. If the slip lands face down, folded over, out of reach, or in the way of the row, lift it clear with the **right gripper** and lay it again.

Step 4: Stage the three items

Goal: the carton empty and all three items standing in one row in slip line order, the third one handed from the right gripper to the left gripper on its way there.

Every item is set down in the row by the left gripper. Lines 1 and 2 are packed in the carton's left half, in the left arm's reach, so the left gripper lifts each one out and stages it with no pass. Line 3 is packed in the right half, too far out for the left arm to reach cleanly, so the right gripper lifts that one as far as the handoff point and passes it across. The right gripper reads every line and never sets an item down on the desk. The read comes first every time, and no gripper goes to the carton until the read is done.

4.1 Read line 1 and stage its item

- The **right gripper**, closed and empty, travels to the **slip spot** and holds still over **line 1** for 2 seconds, then lifts clear.
- The **left gripper** closes on the item that line names, in the carton's left half, at its grasp, and lifts it straight up out of the carton until there is daylight under it.

- Carry it level to the **outer left end** of the **row band**, out past where the other two will go, lower it until it is resting flat on the desk, and release. This item sets where the row starts.
- Let it settle before the next line.

4.2 Read line 2 and stage its item

- The **right gripper** reads **line 2** the same way and lifts clear.
- The **left gripper** lifts the item that line names straight up the same way and stages it at the **next spot** to the right of item 1, one step back toward the carton, lowered straight down and left with a clear gap to its neighbor.
- Let it settle before the next line.

4.3 Read line 3, pass it across, and stage it

Line 3 is the one item the right gripper lifts, and the right gripper never carries it past the handoff point. It still lands in the row by the left gripper, like the other two.

- The **right gripper**, closed and empty, holds still over **line 3** at the **slip spot** for 2 seconds, then lifts clear.
- The **right gripper** closes on the item that line names, in the carton's right half, at its **grasp**, and lifts it straight up out of the carton until there is daylight under it.
- Carry it level to the **handoff point** in front of the carton, in the orientation it was packed in, and hold still there with the item level.
- The **left gripper**, open and empty, comes to the item at the handoff point and closes on its **cross grasp**, at the same middle and at right angles to the hold the right gripper already has.
- Once the **left gripper is closed** on the item, the **right gripper** opens, lifts clear, and goes back to its own side of the desk. The item stays level and still through the pass. It is not turned, tilted, shifted, or set down.
- The **left gripper** carries it level to the **next spot** to the right of item 2, the one nearest the carton, lowers it until it is resting flat on the desk, and releases.
- The gripper lifts clear and the item is left to settle, with a clear gap off the carton's left wall.

Take each item straight up. Never slide one out from between its neighbors, never move one item aside to reach another, and never nudge a placed item to make room for the next one. Leave the gap when you set the item down. The left gripper carries out to the left of the carton and never over it.

In the pass, the left gripper closes before the right gripper opens, never the other way round: the item is held by one gripper or by both, never by neither. Never tip or tilt the carton to bring the item within reach; if it sits deep in the carton, come down on it from above. Never set the item on the desk partway across to take a new hold, and never let the left gripper reach into the carton's right half for it.

Check: three items stand in one row to the left of the carton in slip line order left to right, item 1 at the outer end and item 3 nearest the carton with a clear gap off its left wall, each resting flat and settled, roughly in line with each other, with a clear gap between neighbors and none touching, each one the same way up it was packed, and the carton floor is bare from front to back. If an item is out of order, lift it clear and lower it in its right place.

Step 5: End the episode

Each line was read before its item came out of the carton, so the row already is the count. The slip is not read again here.

- Confirm three items stand in the row, the carton is empty, all four flaps are still down, the slip lies untouched at the slip spot, and the cutter lies on its rest.
- Return both arms home, clear of the carton, the row, the handoff point, and the slip, then stop recording.

Never reshuffle the row at the end to make it match the slip. A row that came out wrong is tagged and left as it is.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the three items off the desk and lay them back in the carton in one layer, in a single row running left to right, with a clear gap between them, each one the same way up it stood in the row. Nothing rests on top of anything else.
2. Vary which item sits on which line between episodes rather than rebuilding the same order every time, and print a slip whose three lines match the row the items were packed in. Whichever item ends up on **line 3** has to take both grasps at once, so put an item there that can be passed.
3. Lay the slip face up on top of the contents in the carton's right half, where the right gripper takes it.
4. Lift all four flaps off the outside walls. Fold the two side flaps in flat over the opening first, then fold the two end flaps over them so they meet edge to edge along the seam. Flatten the creases so no flap stands proud.
5. Lay a fresh tape strip flat along the seam, covering it end to end, with no tape carried down the walls. Clear the cut strip off the desk.
6. Set the carton square at back center, base flat on the desk, flaps closed and facing up, with clear desk on all four sides.
7. Check the cutter: blade seated, depth set to tape only, and the head parts a strip in one flat draw. A blade that drags, tears, or leaves the tape joined is replaced before the next episode. Lay it on the cutter rest head away from the front edge.

8. Replace anything that is damaged: a carton that will not sit square or whose flaps will not stay creased down, a slip that is torn or no longer readable, or an item that is crushed, split, open, leaking, or will no longer stand or lie still on the desk.
9. Wipe the desk and confirm it is dry and bare. A slick or crumbed desk lets the carton slide under an anchor.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** an end flap is opened before the tape is parted along the whole seam, a side flap is opened before both end flaps are down, the slip is lifted before all four flaps are down, an item is lifted before the slip is resting on the desk, or the lines are staged in any order other than 1 to 3.
- **SOP rule broken:** Steps 1 to 4, the order never changes: cut, flaps, slip, then the three items in slip line order 1 to 3.
- **Coaching note:** finish the step you are in before you start the next one.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes the cutter, the right side flap, the near end flap, or the slip, reaches into the carton's right half for the line 3 item, or is the one held over the slip to read; or the right gripper takes the left side flap, the far end flap, reaches into the carton's left half, enters the row band, or sets an item down on the desk.
- **SOP rule broken:** Steps 1 to 4, each arm keeps its own tool, flap, and half of the carton, and every item in the row is set down by the left gripper.
- **Coaching note:** each arm stays on its own side. The only thing that crosses the middle is the line 3 item, and it crosses in the pass.

Violation: Worked without an anchor

- **Visible cue:** the cutter is on the seam or a flap is being folded while the other gripper is off the carton, or the anchoring gripper lets go before the working gripper is clear.
- **SOP rule broken:** Steps 1 and 2, nothing else holds the carton, so every cut and every flap fold is anchored on the opposite side.
- **Coaching note:** anchor first, let go last.

Violation: Anchor in the wrong place

- **Visible cue:** the anchoring gripper sits on the seam or in the blade's path during the cut, presses on the flap the other gripper is working, presses on the contents or reaches across the open top, crosses onto the other gripper's side, or slides along the wall or the flap during the step.
- **SOP rule broken:** Steps 1 and 2, the anchor sits still on its own side, either against the outside of the wall or down on a flap that is still closed, clear of the seam and of the flap being worked.
- **Coaching note:** wall or closed flap, on your own side, and hold it still.

Violation: Cutter taken by the head

- **Visible cue:** the gripper closes on the head of the cutter, on the blade end, or across the cutter at an angle instead of on the flat of the handle.
- **SOP rule broken:** Step 1, a gripper closes on the flat of the handle only, never on the head.
- **Coaching note:** land on the handle or reset and take it again.

Violation: Blade off the tape strip

- **Visible cue:** the cutter wanders off the seam onto the flap face, is drawn in an arc rather than along the line of the seam, is drawn left end to right end, or scores the carton beside the strip.
- **SOP rule broken:** Step 1, one pass along the seam, right end to left end, and the blade touches nothing but the tape.
- **Coaching note:** put the head on the strip and keep it there for the whole run. Stopping on the strip is fine; wandering off it is not.

Violation: Head lifted or plunged during the cut

- **Visible cue:** the head is lifted off the strip partway along and set down again, or the cutter is driven down into the carton rather than kept flat, and the blade reaches below the tape. A draw that stalls and carries on with the head still flat on the strip is not this violation.
- **SOP rule broken:** Step 1, keep the sole of the head flat on the carton top for the whole draw, stalls included.
- **Coaching note:** flat on, flat all the way, flat off. If it sticks, wait on the strip and keep going.

Violation: Flaps forced against joined tape

- **Visible cue:** an end flap is pulled, levered, or torn open while part of the tape strip is still joined, a side flap is worked at while an end flap still lies over it, or the tape rips rather than parting along the cut.
- **SOP rule broken:** Steps 1 and 2, no flap is torn open against tape that is still joined.
- **Coaching note:** if it resists, put the cutter back on the joined part.

Violation: Cutter not returned to its rest

- **Visible cue:** the cutter is left on the carton, in the row band, on the slip, or still in the gripper when the flaps are opened, or it is dropped toward the desk rather than lowered onto it.
- **SOP rule broken:** Step 1, the cutter is back flat on the cutter rest before any flap is touched.
- **Coaching note:** the tool goes back before the next step starts.

Violation: Flap left over the opening

- **Visible cue:** a flap is left standing up, leaning back across the opening, or resting on the contents, a side flap is left tucked in flat over the opening, or items are lifted out past a flap that is still over the opening.
- **SOP rule broken:** Step 2, all four flaps lie creased down against their own outside walls with the opening clear.
- **Coaching note:** crease it down and hold it before you let go.

Violation: Slip mishandled

- **Visible cue:** the slip is laid face down, lands in the row band, under an item, out of the right arm's reach, or on the cutter rest, is folded or creased, is left in the carton, or is moved, covered, picked up, or pressed after it has been laid down, including by the gripper reading it.
- **SOP rule broken:** Step 3, the slip lies face up and readable at the slip spot and stays untouched for the rest of the episode. A read hovers over the line and never lands on the slip.
- **Coaching note:** face up, out of the way, then leave it alone.

Violation: Line not read before the lift

- **Visible cue:** an item is lifted with no read over its line first, the read comes after the item is already moving or already down, the gripper passes over the slip without stopping on the line, or the reading gripper is holding an item while it reads.
- **SOP rule broken:** Step 4, the right gripper reads the line, closed and empty, and holds over it for 2 seconds before the item on it is touched. These reads are the only count the episode gets.
- **Coaching note:** read the line first, then go for the item. One read, one item.

Violation: Pass skipped or run by the wrong arm

- **Visible cue:** the left gripper reaches into the carton's right half and lifts the line 3 item itself, the right gripper carries that item past the handoff point and sets it down on the desk or in the row, or the item is put down partway across and picked up again instead of being passed in the air. It is also this violation if the right gripper passes an item from the carton's left half that the left gripper should have lifted itself.
- **SOP rule broken:** Step 4, the right gripper lifts line 3 as far as the handoff point and the left gripper is what carries it into the row. Lines 1 and 2 are lifted and staged by the left gripper with no pass.
- **Coaching note:** right hand out of the carton, left hand onto the desk. Only line 3 is passed.

Violation: Bad pass

- **Visible cue:** the right gripper opens before the left gripper has closed on the item, both grippers are off the item at once, the item slips, tips, turns, or is regripped during the pass, the left gripper closes on the same faces the right gripper is holding instead of crossing at right angles, or the pass happens over the carton, the slip, the row, or anywhere but the handoff point.
- **SOP rule broken:** Step 4, the left gripper closes on the cross grasp first and only then does the right gripper open, with the item held level at the handoff point and never turned.
- **Coaching note:** close, then open. The item is never held by neither gripper.

Violation: Row out of slip order

- **Visible cue:** an item lands out of its slip line position, an item is placed further out to the left than one already down, or the row reads in an order the slip does not list.
- **SOP rule broken:** Step 4, the row is built left to right in slip line order from the outer left end back toward the carton, and the row order is the count.
- **Coaching note:** read the line, take the item on it, then put it at the carton end of the row.

Violation: Items crowded or touching in the row

- **Visible cue:** two staged items touch, lean on each other, or sit stacked, an item is set down so close that the next one has nowhere to go, or the row is left as a cluster rather than a line.
- **SOP rule broken:** Step 4, a clear gap between neighbors and nothing touching.
- **Coaching note:** leave the gap when you set it down, not afterwards.

Violation: Placed item moved to make room

- **Visible cue:** a gripper nudges, slides, or lifts an item that is already in the row to fit the next one in, or the row is shuffled along the desk.
- **SOP rule broken:** Step 4, never nudge a placed item to make room for the next one.

- **Coaching note:** the row grows to the right. Nothing already down gets touched.

Violation: Item dug out from between its neighbors

- **Visible cue:** an item is slid, pulled, or worked out from between the items beside it, or a gripper moves one item aside to reach another.
- **SOP rule broken:** Step 4, no item is pulled out from between its neighbors and each item is taken straight up.
- **Coaching note:** everything is in one layer already. Come down on it and lift straight up.

Violation: Item taken by the wrong part

- **Visible cue:** an item is taken by a lid, a cap, a corner, an edge, a flap, or a loose part rather than at its middle, or the slip is taken by a corner rather than its near edge.
- **SOP rule broken:** Steps 3 and 4, each item has one grasp and it is taken the same way every episode.
- **Coaching note:** land on the grasp point or reset and take it again.

Violation: Item dragged instead of lifted

- **Visible cue:** an item slides along the carton floor, the carton wall, or the desk still touching it, with no daylight under it in transit.
- **SOP rule broken:** Steps 3 and 4, lift straight up until there is daylight before moving sideways.
- **Coaching note:** the lift is its own motion, not the start of the carry.

Violation: Item turned in the air

- **Visible cue:** an item tips, rolls, is flipped, or is swung off the way up it was packed during the carry or during the pass, or the slip is turned over between the carton and the desk.
- **SOP rule broken:** Steps 3 and 4, nothing is turned in the air and each item is carried in the orientation it will be set down in.
- **Coaching note:** orientation is set at the pick, never in the air.

Violation: Regripped in the air

- **Visible cue:** the gripper shifts, rolls, or seats its hold again on an item or on the cutter without setting it down first.
- **SOP rule broken:** Steps 1 to 4, set it down before taking a new hold. The pass in Step 4 is not a regrip: the hold moves from one gripper to the other, and neither gripper reseats its own hold.
- **Coaching note:** put it down before you take a new hold. Passing is not regripping.

Violation: Item dropped instead of lowered

- **Visible cue:** a gripper opens above the desk and the item falls the rest of the way, or an item is set down short of its spot and slid across the desk into place rather than lowered straight down onto it.
- **SOP rule broken:** Steps 1 to 4, release only once the item is resting.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the carton, over the slip, over another item, or over open space, and the item lands anywhere but at the spot it was going to.
- **SOP rule broken:** Steps 1 to 4, a gripper opens over the spot the item is going to and nowhere else. The right gripper opening at the handoff point, after the left gripper has closed, is not this violation.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Contents pressed or packed down

- **Visible cue:** a gripper presses, pats, or squashes an item in the row, the slip on the desk, or the items still in the carton, or holds an item down against the desk after it is already resting.
- **SOP rule broken:** Step 4, each item is lowered until it is resting and nothing is pressed down. The 2 second press in Step 2 is for the flap crease only.
- **Coaching note:** leave it how it lands.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, or slides an item into position after releasing it, instead of lifting it clear and lowering it again.
- **SOP rule broken:** Steps 3 and 4, correct a bad placement by lifting the item clear and lowering it into place again.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Item dropped, collided, or knocked something over

- **Visible cue:** an item or the cutter falls onto the desk short of its spot or onto the floor, the carton is struck and shifts, an already staged item is knocked over or out of the row, or the two arms strike each other, including at the handoff point.
- **SOP rule broken:** Steps 1 to 4, keep each item on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with something still in the carton, an item missing from the row, the row out of slip order, a flap back over the opening, the cutter off its rest, an arm

away from home, or an arm still over the carton, the row, the handoff point, or the slip. It is also a violation to rearrange the row at the end to make it match the slip.

- **SOP rule broken:** Step 5, confirm the row, the empty carton, and the flaps, return both arms home clear of the scene, then stop recording.
- **Coaching note:** confirm, hold, then home. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Blade will not part the tape** under a flat draw along the strip, or the blade is missing, dull, or set to the wrong depth. Replace the cutter before the next episode.
- **Tape strip missing, already split, or wrapped down the walls** when the episode begins.
- **Packing slip missing, blank, or unreadable** when the step begins.
- **Count mismatch:** the carton holds fewer or more items than the slip lists, or an item in the carton is not on the slip. Discard the episode and restock the carton to match the slip.
- **Carton slides under a correct anchor** because the desk is slick, wet, or crumbed. Wipe and dry the desk before the next episode.
- **Carton will not sit square** on the desk, or a flap will not hold its crease and springs back over the opening under its own weight. Replace the carton before the next episode.
- **Defective item:** an item that is crushed, split, open, leaking, or will not stand or lie still on the desk in the orientation it is packed in, or a line 3 item that cannot be held on both grasps at once so it can be passed. Replace it before the next episode.

Annotation subtasks (from SOP)

1. Anchor the carton
2. Take the safety cutter from its rest
3. Draw the cutter along the tape seam
4. Return the safety cutter to its rest
5. Fold one flap down against its outside wall
6. Lay the packing slip on the desk
7. Read a line on the packing slip
8. Lift one item out of the carton
9. Pass an item from the right gripper to the left gripper
10. Place one item in the row

11. End the episode